IN5340/IN9340 Project III

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Code: `github.com/Twaal/Statistical_Signal_Processing_IN5340/tree/master/Project_3`

Agenda

Problem statement

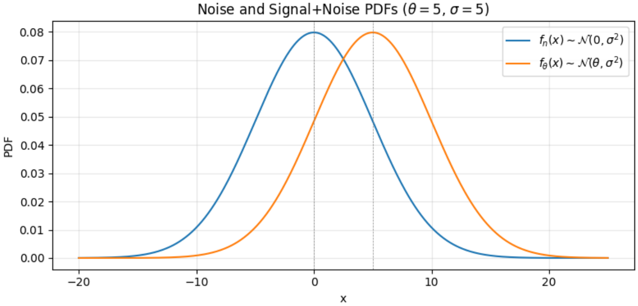
- Binary detection with additive Gaussian noise:

$$H_0 : x = n, \qquad H_1 : x = \theta + n, \quad n \sim \mathcal{N}(0, \sigma^2)$$

- Use $\theta = 5$ and $\sigma = 5$ as baseline case.
- Design and evaluate:
  - CFAR detector for given P_{FA}
  - ROC curve (theoretical vs empirical)
  - Improvement using sample mean of N measurements

b) Noise and signal+noise PDFs

- Under H_0: $x \sim \mathcal{N}(0, \sigma^2)$
- Under H_1: $x \sim \mathcal{N}(\theta, \sigma^2)$
- For $\theta/\sigma = 1$, distributions overlap strongly.
- Detection is possible, but not highly reliable in this regime.

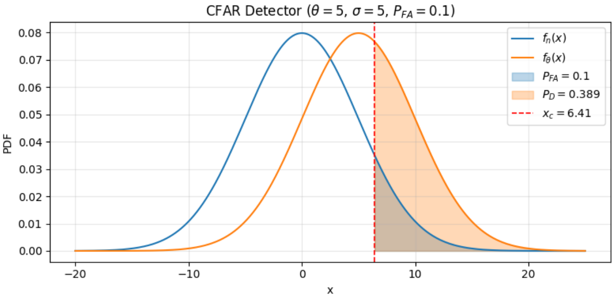


c) CFAR detector with $P_{FA} = 0.1$

$$x_c = \sigma Q^{-1}(P_{FA}) = 5\, Q^{-1}(0.1) \approx 6.41$$

$$P_D = Q\left(\frac{x_c - \theta}{\sigma}\right) \approx 0.389$$

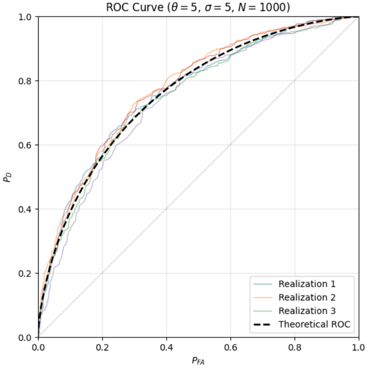
- Blue shaded area: false-alarm probability P_{FA}.
- Orange shaded area: detection probability P_D.
- Dashed red line: threshold x_c.



- Data model:

  $s \in \{0, \theta\},\ P(s=0) = P(s=\theta) = 0.5,\ x$

- Empirical ROC computed by threshold sweep on simulated data.
- Multiple realizations vary, but cluster around the theoretical ROC.



e) Multi-sample detector and minimum N

Using sample mean of N measurements:

$$\bar{x}|H_0 \sim \mathcal{N}\left(0, \frac{\sigma^2}{N}\right), \qquad \bar{x}|H_1 \sim \mathcal{N}\left(\theta, \frac{\sigma^2}{N}\right)$$

For CFAR with requirements $P_{FA} = 0.01$ and $P_D > 0.9$:

$$N \geq \left(\frac{\sigma}{\theta}\right)^2 \left(Q^{-1}(P_{FA}) + Q^{-1}(1 - P_D)\right)^2$$

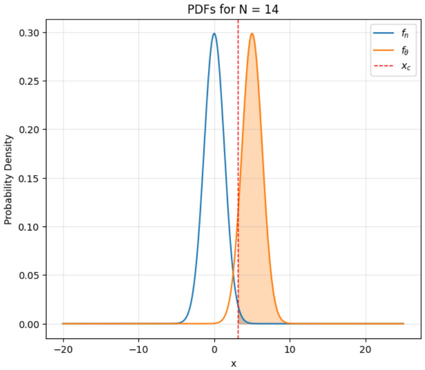
Numerically:

$$N_{\text{min, cont}} \approx 13.02 \quad \Rightarrow \quad N_{\text{min}} = 14$$

At $N = 14$: $x_c \approx 3.11$ and $P_D \approx 0.922$ (requirement satisfied).

f) PDFs for improved detector ($N = 14$)

- Variance shrinks from σ^2 to σ^2/N.
- The two PDFs become better separated.
- This improves detection performance for fixed P_{FA}.



Summary

- For single measurement ($\theta = \sigma = 5$), overlap is substantial and P_D is modest at practical P_{FA}.
- CFAR with $P_{FA} = 0.1$ gives threshold $x_c \approx 6.41$ and $P_D \approx 0.389$.
- ROC simulations agree with theoretical ROC, with expected realization-to-realization variation.
- Averaging multiple measurements improves separability; minimum requirement is met at $N = 14$.